

# **Porphyrin-Containing Metallacage with Precise Active Sites and Super Long-Term Stability as a Specific Peroxidase Mimic for Versatile Analyte Determination**

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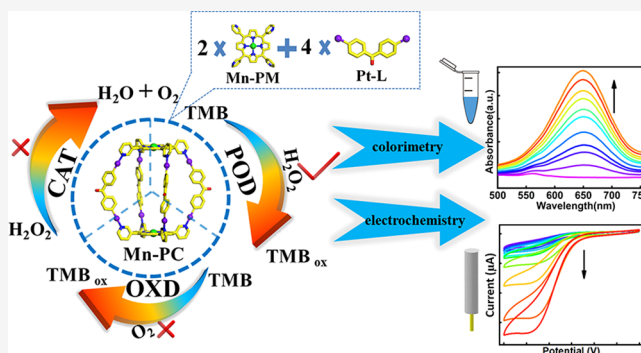
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**ABSTRACT:** Inspired by the architecture of single-atom catalysts, where the monodispersed metal atoms are widely distributed but stabilized by various coordination circumstances, the biomimetic design and synthesis of metalloporphyrin-containing nanocages have been demonstrated in this study. The nanocages were fabricated through a coordination-driven self-assembly process, and the Mn(III) porphyrin-based one was found to have exclusively peroxidase-like activity at pH 6.0 with neither oxidase nor catalase-like activity under the routine conditions. Benefiting from this, we demonstrated the wide applicability and convenient usage of an Mn(III)-containing supramolecular nanocage (Mn-PC) in the one-step detection of  $\text{H}_2\text{O}_2$ , sarcosine, and glucose through various oxidase-involved reactions, with a satisfactory detection limit and eligible specificity. Real samples including  $\text{H}_2\text{O}_2$  in lens care solution, sarcosine in human urine, and glucose in human serum were also assayed, showing an adequate recovery rate. Such a specific activity originates from the super-consistent microstructure of each catalytic unit, which means that the active site of manganese porphyrin was “protected” by the confinement of the nanocage. This also helps to sustain the super long-term activity even after 545 days of storage. Furthermore, the intrinsic electronic structure of the Mn(III)-containing supramolecular nanocage endows the ability in electrochemical detection of  $\text{H}_2\text{O}_2$  and glucose. Our smart design toward the supramolecular nanocages with a defined structure and quantity contributes to the construction of the ingenious sensing platform and has guiding significance for architectural design of nanozymes.



Nanozymes generally refer to the artificial nanomaterials intrinsically with enzyme-like potency whose ascendancy over their natural counterparts is embodied in low-cost, durable performance under extreme conditions, effortless mass production, sustained storage, and most importantly, the expediency of activity modulation.<sup>1,2</sup> So far, various nanomaterials or macromolecules with enzyme-like activities have been reported, including noble metals,<sup>3,4</sup> metal oxides,<sup>5–7</sup> carbon-based nanomaterials,<sup>8</sup> and porphyrin derivatives.<sup>9,10</sup> Although the research on nanozymes has made great progress, the inhomogeneous elemental distribution and low density of active sites of most reported nanozymes result in lower activity and poorer specificity compared to natural enzymes, and the various facet structures on the surface complicate the investigation on the origin for the enzyme-like activity of the nanozymes.<sup>11</sup> Another issue that severely compromises the activities is the low utilization efficiency of the metal atoms.<sup>12</sup> Single-atom nanozymes (SANs), which maximally expose and utilize the atomically dispersed active sites, have potential in achieving high catalytical activity and specificity like natural enzymes and are conducive to unveiling the structure–activity relationship.<sup>13,14</sup> However, for SANs, the synthesis mostly goes

through a high-temperature pyrolysis process, and the characterization techniques like synchrotron radiation and spherical aberration electron microscopy are both time- and energy-consuming processes.<sup>15–17</sup> Additionally, the structural identification for SANs remains a critical challenge as the stabilizing ligand for the metals could appear in various forms,<sup>18</sup> like the case with metal organic frameworks (MOFs), although also reported to have enzymatic activity with a high surface area and abundant active sites.<sup>19–21</sup> It leads to frustration in the evaluation of the catalytic constant ( $K_{\text{cat}}$ ), which is regarded as an important parameter for enzymatic reactions.<sup>22</sup>

Metalloporphyrins are commonly used for the biomimetic design of artificial enzymes because they are illustrious embodiments exerting biological functions for many enzyme

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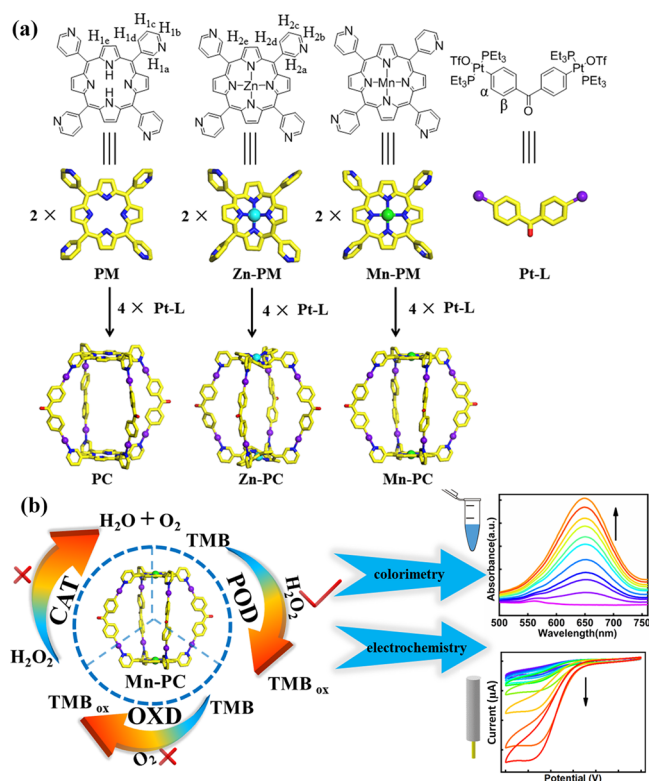
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families. For example, hemin, an iron(III)-binding porphyrin, is well known as an imitation of the active center for horseradish peroxidase (HRP). However, hemin has a conspicuous propensity of aggregating in the water phase, greatly inhibiting the catalytic performance.<sup>10,23</sup> People have made many efforts to prevent the aggregation of metalloporphyrins, like immobilizing the metalloporphyrin catalysts on various supports,<sup>24–27</sup> but the products are heterogeneous, as stated above, lacking definite molecular structures. Taking the above aspects in mind, and enlightened by the fact that metalloporphyrins are active in enzymatic reactions, supramolecules containing metalloporphyrin structures were considered to possibly have similar behaviors.<sup>28</sup>

In this study, based on the fact that coordinated metal ions with variable valence states usually function as the enzymes' active center,<sup>29</sup> we synthesized the Mn(III)-containing supramolecular nanocage named Mn-PC (Scheme 1a) on the basis

**Scheme 1. Schematic Illustration for (a) Self-assembly of Complexes PC, Zn-PC, and Mn-PC and (b) Colorimetric and Electrochemical Biosensing Based on Mn-PC**



of the coordination-driven self-assembly methodology, where the enzyme-like activity is more prominent compared with the porphyrin cage (PC) and zinc porphyrin cage (Zn-PC) reported previously.<sup>30</sup> As a simple assembly, such a supramolecular nanocage possesses a highly ordered structure. Hence, the molar concentration of nanozymes, as well as the catalytically active center, could be demonstrably figured out. Each nanocage works as an independent catalyst unit, inheriting the merits of MOF-based nanozymes and SANs, meanwhile exhibiting the special superiority by the high yield and easy characterization. The cage-type structure accommodates the active sites in a super-consistent microenvironment, as the state-of-art complicated skeleton of the natural enzymes. Thus, Mn-PC exclusively possesses peroxidase

(POD)-mimicking activity, without oxidase (OXD)-like or catalase (CAT)-like characteristics.

It has been widely recognized that the level of some bioactive molecules has extraordinary significance in terms of early diagnosis.<sup>31</sup> For example, sarcosine levels in urine or blood can evince the situation of prostate cancer.<sup>32,33</sup> Hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>), a hazardous chemical agent as announced by the World Health Organization,<sup>34</sup> can oxidize the chromogenic substrate under the catalysis of POD and is also an intermediate product of various oxidase-involved reactions. Thus, a POD mimic can be employed in analyzing H<sub>2</sub>O<sub>2</sub> or, more usually, in combination with oxidases producing H<sub>2</sub>O<sub>2</sub> to detect the bioactive molecules. Herein, the spectrochemical detection of H<sub>2</sub>O<sub>2</sub>, sarcosine, and glucose was developed through a one-step strategy in a mild acid environment, exhibiting an eligible detection limit. Moreover, the electrochemical detection of H<sub>2</sub>O<sub>2</sub> and glucose was also realized owing to the intrinsic electronic structure of Mn-PC (Scheme 1b).

## EXPERIMENTAL SECTION

Materials and instruments, synthesis process, the POD, OXD, and CAT-like activities of PC, Zn-PC, and Mn-PC are in the Supporting Information.

**Catalytic Mechanism Investigation.** In 10 mM pH 6.0 NaAc-HAc buffer solution, H<sub>2</sub>O<sub>2</sub>, 3,3',5,5'-tetramethylbenzidine (TMB), and Mn-PC were added to make final concentrations of 2.5 mM, 0.8 mM, and 50 μg/mL, respectively, followed by the addition of thiourea solution with various concentrations. After a 40 min reaction at 25 °C, the absorbance of TMB<sub>ox</sub> at 652 nm was detected using a Tecan Spark microplate reader (Tecan, Männedorf, Switzerland). The detailed experimental procedures with 5,5-dimethyl-1-pyrroline N-oxide (DMPO) as a radical scavenger are as follows: In 10 mM pH 6.0 NaAc-HAc buffer solution, Mn-PC, DMPO, and H<sub>2</sub>O<sub>2</sub> were added in sequence to make final concentrations of 50 μg/mL, 100 mM, and 5 mM, respectively, to make a homogeneous mixture that was transferred to a paramagnetic tube for electron spin resonance (ESR) detection.

**Kinetic Study on the Enzymatic Reaction.** The kinetic study was performed by fixing the concentration of one substrate while changing the other one. In particular, 5 μL of H<sub>2</sub>O<sub>2</sub> with different concentrations (5, 10, 20, 40, 50 mM) were added to the solution of 10 mM NaAc-HAc (pH 6.0) containing 50 μg/mL of Mn-PC. After that, 5 μL of 1 mM TMB was added to the above solutions, respectively, to make a final solution with a TMB concentration of 0.02 mM. Then, 200 μL of the obtained mixtures were added to a 96-well plate and allowed to react at 25 °C. The absorbance at 652 nm was detected in a kinetic mode at an interval of 1 min in 2 h. According to Lambert–Beer's Law:  $A = \epsilon bc$ ,  $C_{\text{TMB}_{\text{ox}}}$  could be figured out first. Then the reaction rate was calculated according to the formula  $V = C/t$ . According to the Michaelis–Menten equation:  $1/V = K_m/V_m * 1/[S] + 1/V_m$ , the double-reciprocal curves for H<sub>2</sub>O<sub>2</sub> and TMB were drawn. To obtain more similar double-reciprocal curves, the final concentration of H<sub>2</sub>O<sub>2</sub> and TMB can be adjusted to 0.1, 0.2, and 0.4 mM, respectively.

**Colorimetric Detection of H<sub>2</sub>O<sub>2</sub>, Sarcosine, and Glucose.** Colorimetric detection of H<sub>2</sub>O<sub>2</sub> was conducted in 250 μL of sodium acetate buffer (10 mM, pH 6.0) containing 50 μg/mL Mn-PC, 0.8 mM TMB, and different concentrations

of  $\text{H}_2\text{O}_2$ . The reaction was allowed to react at 25 °C for 40 min.

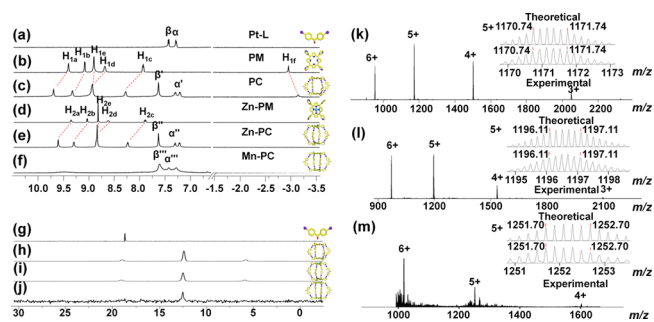
Colorimetric detection of sarcosine via a one-step method was conducted in a buffer mixture (10 mM NaAc-HAc + 10 mM  $\text{NaH}_2\text{PO}_4\text{-Na}_2\text{HPO}_4$ , pH 6.5), which contained 0.4 mg/mL of sarcosine oxidase (SOx), 50  $\mu\text{g/mL}$  of Mn-PC, 0.8 mM TMB, and different concentrations of sarcosine (0–500  $\mu\text{M}$ ). After a 90 min reaction at 30 °C, the absorbance at 652 nm was detected using a microplate reader. In the two-step method detection of sarcosine, 0.4 mg/mL of SOx and different concentrations of sarcosine (0–500  $\mu\text{M}$ ) were incubated in 200  $\mu\text{L}$  of  $\text{NaH}_2\text{PO}_4\text{-Na}_2\text{HPO}_4$  buffer solution (10 mM, pH 6.5) at 37 °C for 90 min. Then 300  $\mu\text{L}$  of NaAc-HAc buffer solution (10 mM, pH 6.0) containing 50  $\mu\text{g/mL}$  of Mn-PC and 0.8 mM TMB was added to the above solution and incubated for another 40 min at 25 °C before absorbance measurement.

Similarly, colorimetric detection of glucose using a one-step method was conducted as follows: 1 mg/mL of GOD, 50  $\mu\text{g/mL}$  of Mn-PC, 0.8 mM TMB, and different concentrations of glucose were incubated in 10 mM NaAc-HAc (pH 6.0) buffer solution at 30 °C for 40 min. In the two-step method detection of glucose, 1 mg/mL of GOD with different concentrations of glucose were first incubated in 10 mM NaAc-HAc (pH 6.0) buffer solution at 37 °C for 30 min. Then 12.5  $\mu\text{L}$  of Mn-PC (1 mg/mL) and 10  $\mu\text{L}$  of TMB (20 mM) were added to the above solution and incubated for another 20 min at 25 °C before absorbance measurement.

**Electrochemical Detection of  $\text{H}_2\text{O}_2$  and Glucose.** The glassy carbon electrodes (GCEs, 3.0 mm in diameter) were first polished successively with 1.0 and 0.3  $\mu\text{m}$  alumina slurry and rinsed with deionized water, followed by sonication in water for 1 min. Then, the electrode was blown-dried with nitrogen. Afterward, 6  $\mu\text{L}$  of Mn-PC (1 mg/mL containing 0.2% Nafion) was drop-cast on the surface of the GCE and dried at room temperature (denoted as Mn-PC/GCE). After drying, 10  $\mu\text{L}$  of GOD (10 mg/mL containing 0.2% Nafion) was then drop-cast on the surface (denoted as Mn-PC@GOD/GCE) and dried at 4 °C. Then, the modified electrode was activated by successive cyclic potential scans at a scan rate of 50 mV/s in 10 mL 10 mM PBS containing 0.1 M KCl (pH 7.0) until steady voltammograms were obtained. The three-electrode system was composed of a GCE (working electrode), an Ag/AgCl electrode (reference electrode), and a Pt wire electrode (counter electrode).

## RESULTS AND DISCUSSION

**Synthesis and Characterization of PC, Zn-PC, and Mn-PC.** The detailed synthesis strategy of PC, Zn-PC, and Mn-PC is illustrated in Scheme S1 (in the Supporting Information (SI)). Based on the coordination-driven self-assembly approach, 5,10,15,20-tetra(3-pyridyl) porphine (PM), zinc 5,10,15,20-tetra(3-pyridyl)-21H,23H-porphine (Zn-PM), or manganese 5,10,15,20-tetra(3-pyridyl)-21H,23H-porphine (Mn-PM) was mixed with 120° diplatinum(II) ligand (Pt-L) in DMSO at 80 °C for 10 h to afford metallacages PC, Zn-PC, and Mn-PC, respectively. The proton nuclear magnetic resonance ( $^1\text{H}$  NMR) spectra of the three ligands (Pt-L, PM, and Zn-PM) and three cages (PC, Zn-PC, and Mn-PC) are shown in Figure 1a–f. For PC and Zn-PC, because of the coordination of nitrogen atoms and the Pt(II) centers, the signals of the protons on pyridyl groups were shifted downfield in the  $^1\text{H}$  NMR spectra (Figure 1c,e).

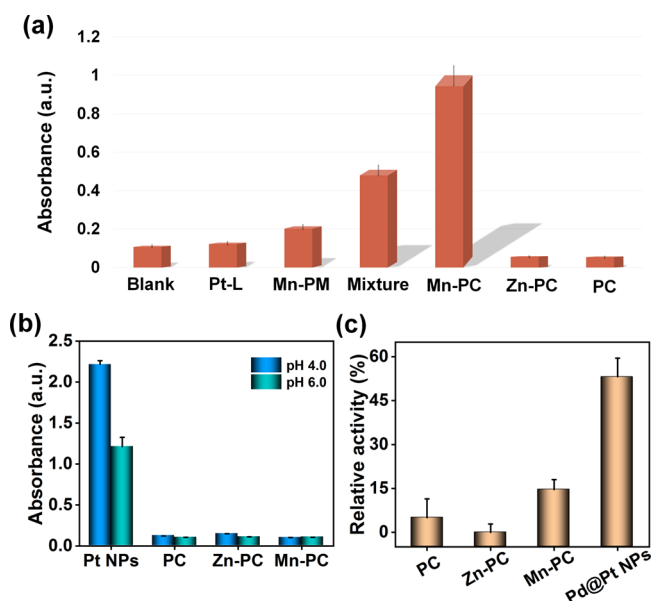


**Figure 1.** (a–f)  $^1\text{H}$  NMR spectra (500 MHz,  $\text{DMSO-d}_6$ , 300 K) of Pt-L, PM, PC, Zn-PM, Zn-PC, and Mn-PC, respectively. (g–j)  $^{31}\text{P}$  NMR (500 MHz,  $\text{DMSO-d}_6$ , 300 K) of Pt-L and three nanocages. (k–m) Electrospray ionization mass spectrometry (ESI-MS) spectra of three nanocages.

At the same time, the signals of pyrrole in PC shifted upfield.<sup>35</sup> However, the paramagnetic nature of the Mn(III) ion causes pronounced effects in NMR spectra (Figure 1f). All the  $^{31}\text{P}$  NMR spectra of PC, Zn-PC, and Mn-PC exhibited a single peak with concomitant  $^{195}\text{Pt}$  satellites ( $\delta = 12.63$  ppm for PC, 12.73 ppm for Zn-PC, and 12.75 ppm for Mn-PC, respectively) (Figure 1g–j), suggesting that a single component was obtained.<sup>36</sup> Compared with Pt-L, these peaks show an upfield shift by approximately 6.29, 6.18, and 6.17 ppm for PC, Zn-PC, and Mn-PC, respectively.

ESI-MS has been recognized as an effective method to determine the molecular formation of obtained assemblies.<sup>37</sup> As shown in Figure 1k–m, all the ESI-MS spectra of PC, Zn-PC, and Mn-PC exhibit one prominent set of peaks with different charge states, because of the loss of different numbers of the trifluoromethanesulfonate counterions.<sup>38</sup> After the deconvolution, the molecular weight of these three metallacages was 6597.5 Da (PC), 6721.3 Da (Zn-PC), and 6821.3 Da (Mn-PC), respectively, matching well with their expected chemical compositions of two porphyrins and four diplatinum(II) ligands (Pt-L). The characterization by NMR and ESI-MS (Figures S1–S17) can readily ensure the formation of all the products, not that time- and energy-consuming compared with the synchrotron characterization. The high-resolution transmission electron microscopy (HRTEM) image of Mn-PC showed well-dispersed nanoparticles (NPs) in the size of  $\sim 2.5$  nm (as roughly outlined by the dotted frames in Figure S18), consistent with 2.2 nm of PC obtained from the 2D diffusion-ordered spectroscopy (DOSY) spectrum (Figure S6).

**Explanation of the Specificity of POD-like Activity of Mn-PC.** In comparison with PC, Zn-PC, Pt-L, and Mn-PM (the building blocks of Mn-PC), and the mixture of Pt-L and Mn-PM, only Mn-PC exhibits apparent POD-like activity although the concentration was half that of PC and Zn-PC (Figure 2a). The possible reasons are: (i) PC does not have metal atoms with changeable valence state; (ii) the Zn atom in Zn-PC is redox-inert, just like the displacement of Mn(II) with Zn(II) generated the ineffectiveness in antioxygenation;<sup>39</sup> and (iii) the nanocage structure has the ingenuity for metal-porphyrin complexes in their enzyme-mimic function. The smartness of this characteristic structure of Mn-PC is also witnessed by the specificity as POD mimics (Figure 2b,c) when compared with Pt NPs and Pd@Pt NPs, which have OXD-<sup>4</sup> and CAT-like<sup>40</sup> activities, respectively. This is in sharp contrast with the single-atom catalysts, such as Fe-N/C that concomitantly exhibited POD-, OXD-, CAT-, and glutathione

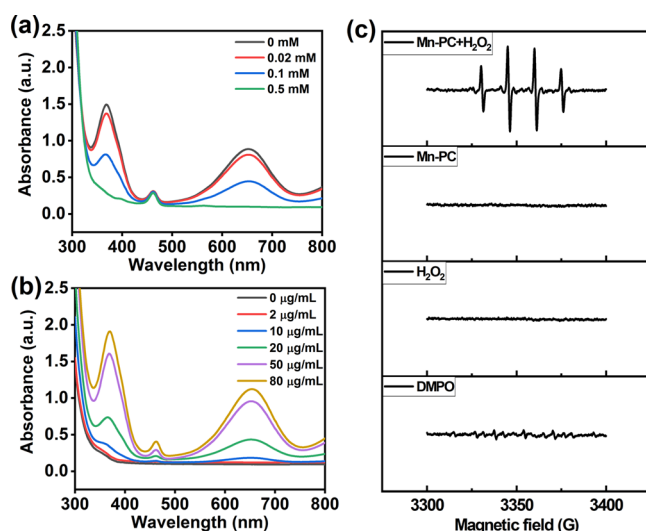


**Figure 2.** (a) Comparison of the POD-like activities of PC, Zn-PC, Mn-PC, Mn-PM, Pt-L, and their mixture. The evaluation of the OXD-like (b) and CAT-like (c) activities of PC, Zn-PC, and Mn-PC. Pt nanoparticles (Pt NPs) or Pd@Pt nanoparticles (Pd@Pt NPs) are used as the positive control.

peroxidase-like activity,<sup>41</sup> as well as the atomically dispersed Mn that was found to have disparate performance in catalyzing oxygen reduction<sup>42</sup> and water oxidation,<sup>43</sup> possibly because of the nuance in the coordination environment. In addition, the specific POD-like activity of Mn-PC was also reflected on the catalytic oxidation of other chromogenic substrates, such as OPD and ABTS (S19).

The specific POD-like activity frees the interference of O<sub>2</sub> from directly oxidizing the substrate to its chromogenic product by O<sub>2</sub><sup>44</sup> and meanwhile reveals the structure–specificity relationships which are helpful for the design of nanozymes. Notably, in the absence of TMB, the incubation of Mn-PC with H<sub>2</sub>O<sub>2</sub> can lead to degradation of the porphyrin ligand periphery, decreasing the intrinsic absorbance at 462 nm for Mn-PC (Figure S20a). This also explains the decline in absorbance at 240 nm (*A*<sub>240</sub> nm) by 14.5% caused by Mn-PC (Figure 2c). Nonetheless, Mn-PC was “protected” by TMB, which will react with H<sub>2</sub>O<sub>2</sub> and the absorbance at 462 nm for Mn-PC will be retrieved (Figure S20b). A similar occurrence was observed for Mn-PM (Figure S20c).

**Investigation of the Catalytic Mechanism of Mn-PC as POD Mimics.** To verify whether the reaction is implicated in the reactive oxygen species (ROS), thiourea was used as the scavenger of ROS.<sup>45,46</sup> The presence of thiourea can diminish the characteristic absorbance of the reaction mixture of TMB, H<sub>2</sub>O<sub>2</sub>, and Mn-PC, which is gradually weakened as the concentration of thiourea increases and completely erased under 0.5 mM thiourea, indicating the elimination of the produced ROS by thiourea (Figure 3a). In addition, the generation of ROS is in a concentration-dependent manner, as the absorbance is strengthened with the increased concentration of Mn-PC under a fixed thiourea concentration of 0.04 mM (Figure 3b) and 0.1 mM (Figure S21). To identify the category of the ROS, ESR spectroscopy was employed, and DMPO was adopted as the trapper for possible production of hydroxyl radicals (•OH). DMPO can react with •OH to yield



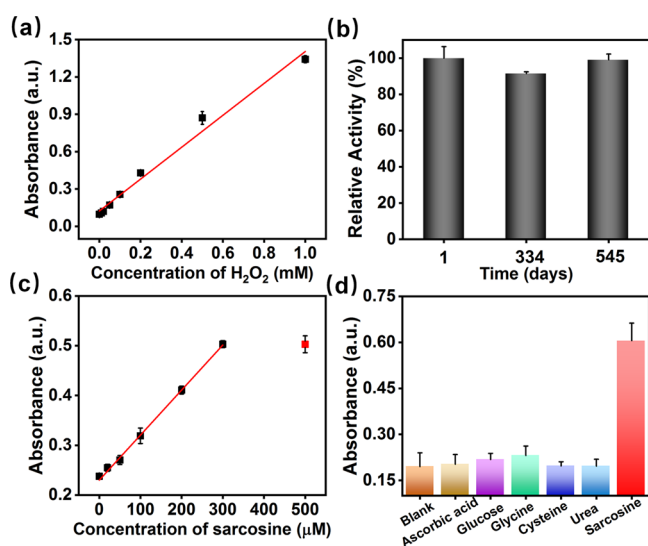
**Figure 3.** (a) UV–vis spectra of the solution after the reaction of 0.8 mM TMB, 2.5 mM H<sub>2</sub>O<sub>2</sub>, 50 μg/mL of Mn-PC, and different concentrations of thiourea (0–0.5 mM) in 10 mM buffer (pH 6.0); (b) UV–vis spectra of the solution after the reaction of 0.8 mM TMB, 2.5 mM H<sub>2</sub>O<sub>2</sub>, 0.04 mM thiourea, and different concentrations of Mn-PC (0–80 μg/mL) in 10 mM buffer (pH 6.0); (c) ESR spectra obtained from the samples including Mn-PC + H<sub>2</sub>O<sub>2</sub> + DMPO and the individual component, respectively.

a hydroxyl radical adduct of DMPO/•OH, which can be evidenced by the characteristic quartette peak patterns with an intensity ratio of 1:2:2:1, as shown in Figure 3c, when incubated with Mn-PC and H<sub>2</sub>O<sub>2</sub>. Hence, the •OH-involved mechanism can be ensured.

**Optimization of Catalytic Reaction Conditions.** In optimization of pH and temperature (Figure S22a,b), the optimums were found to be pH = 6 at 25 °C. When pH is below 6, the protonated TMB<sup>47</sup> has electrostatic repulsion with Mn-PC, and this protonation is weakened as the pH increases, which can be verified by the zeta potential test (Figure S23). We adopted the temperature of 25 °C in all the subsequent experiments if not specifically mentioned. The reaction reaches its plateau after 40 min (Figure S22c). The chromogenic oxidation of TMB was utilized to optimize the concentrations of the reactants, of which the concentration of Mn-PC was first optimized with fixed concentrations of TMB and H<sub>2</sub>O<sub>2</sub>. The maximum absorbance value appears at a concentration of 50 μg/mL (Figure S22d). Similarly, the steady-state kinetics were analyzed by varying the concentration of either TMB or H<sub>2</sub>O<sub>2</sub>, while keeping the other unchanged (Figure S22e,f). The Michaelis constant (*K*<sub>m</sub>), affinity of the enzyme to the substrate, and the maximum reaction rate (*V*<sub>m</sub>) were calculated according to the double-reciprocal plots (Figure S22g,h) and listed in Table S1. Mn-PC has a comparable or even lower *K*<sub>m</sub> value than other artificial enzymes, including atomically dispersed zinc<sup>48</sup> or iron<sup>49</sup> nanozymes. When considering H<sub>2</sub>O<sub>2</sub> as the substrate, the *V*<sub>m</sub> was also comparable to that of zinc-based porphyrin-like structure,<sup>48</sup> Fe–N–C<sup>49</sup> and even higher than that of HRP,<sup>50</sup> suggesting the eligibility of Mn-PC as a POD mimic. Additionally, we measured its activity over a range of TMB and H<sub>2</sub>O<sub>2</sub> concentration (0.1, 0.2, and 0.4 mM), and the slopes of double-reciprocal plots were parallel for two substrates, which is the characteristic of a ping-pong mechanism (Figure S24a,b). The *K*<sub>cat</sub> value generally represents the maximum

turnover rate for individual enzyme active sites. Given the fact that the NP-based nanozymes have plenty of active sites or metal atoms, the  $K_{\text{cat}}$  for every single active site or metal atom should be more reasonable in assessing the activity of the active site or from the perspective of the atom economy. In this study, the value can be steadily acquired by refined calculation, instead of the estimation of the NP-based nanozymes relying on the NP-tracking analysis system that requires NPs to be monodispersed.<sup>22</sup>

**Determination of  $\text{H}_2\text{O}_2$  and Sarcosine.**  $\text{H}_2\text{O}_2$  and sarcosine were chosen as the target models, the detection of which was facilitated by using TMB as a chromogenic substrate. As demonstrated above, pH = 6, 25 °C as the optimal conditions for Mn-PC, was adopted in  $\text{H}_2\text{O}_2$  detection, and a detection range of 0.01–1.0 mM and a detection limit of 1.9  $\mu\text{M}$  with a signal-to-noise ratio of 3 were obtained (Figures 4a and S25). Real samples of  $\text{H}_2\text{O}_2$  in lens



**Figure 4.** (a) Linear correlation between absorbance at 652 nm and  $\text{H}_2\text{O}_2$  concentration. (b) Stability of Mn-PC for 2.5 mM  $\text{H}_2\text{O}_2$  detection. (c) Linear range and (d) selectivity of sarcosine detection.

care solution were also determined by this assay. It can be seen in Table S2 that the recoveries fall in the range of 95.85–104.6%. What is worth noting is the highly durable catalytic activity of Mn-PC that it can maintain the original performance even after 545 days of storage. Such a feature is manifested by the almost unaltered detection results of  $\text{H}_2\text{O}_2$  (Figure 4b). It is speculated that the outstanding stability originates from the confinement of the nanocage, which is a commonly used approach to regulate the surroundings around the light emitters or dyes so as to avoid quenching resulting from aggregation.<sup>51</sup> In the same manner, the confinement effectively prevents the active manganese porphyrin unit from aggregation; otherwise, the aggregation of Mn-PC itself can easily occur in the reaction medium, resulting in a considerable decrease in catalytic activity.<sup>52</sup>

The sarcosine detection was realized through a  $\text{SOx}$ -involved process, transforming sarcosine into  $\text{H}_2\text{O}_2$ , as well as the oxidation reaction catalyzed by Mn-PC, converting the sarcosine level into a colorimetric signal. To acquire the optimal pH value, different pH was attempted in both processes (Figure S26), and a pH value of 6.5 was determined to be the optimal condition, under which the convenient one-

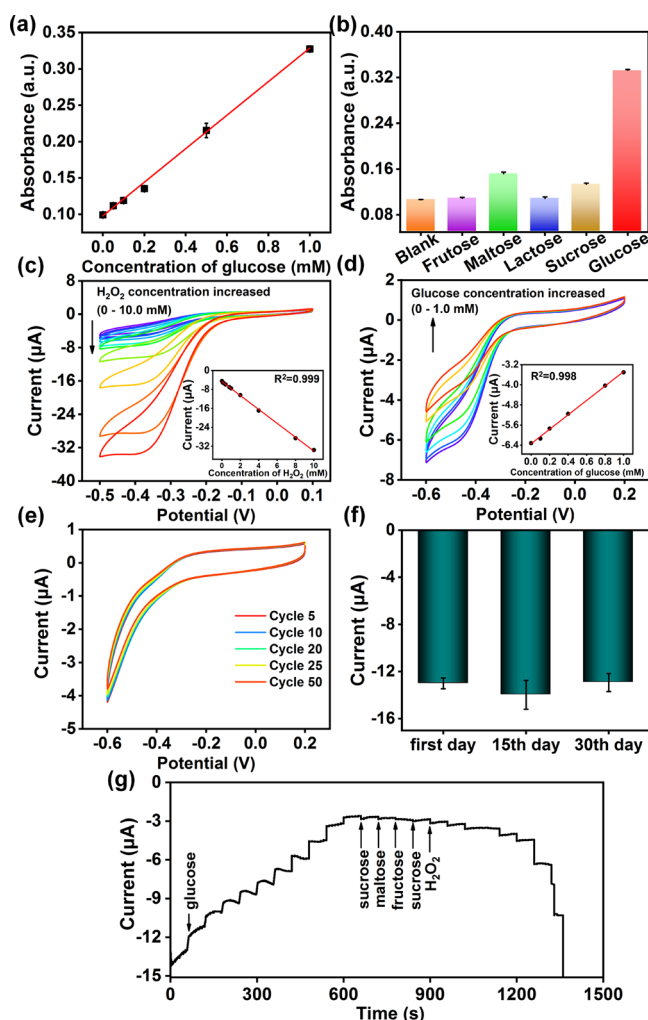
step detection strategy was achieved. A linear relationship between sarcosine concentration and the absorbance values at 652 nm was obtained in the range of 20 to 300  $\mu\text{M}$ , with a detection limit of 2.3  $\mu\text{M}$  (Figure 4c), which is lower than the one of the two-step detection (4.9  $\mu\text{M}$ ) (Figure S27a), suggesting the adequateness of the approach. Several background substances were used to investigate the selectivity of such assay including ascorbic acid, glucose, glycine, cysteine, and urea. As shown in Figure 4d, none of them interfere with the detection process even when they have four times higher concentration (5 mM) than sarcosine (1 mM), indicating the excellent selectivity. To demonstrate the feasibility of this biosensor in practical applications, the analysis of sarcosine was carried out in 4% human urine samples (Table S3). The recovery is located in the range of 100.4–104.8% using the standard addition method, validating that this assay is applicable to determining sarcosine in real samples.

**Universality in Bioanalysis.** Mn-PC has wide applicability in bioanalysis. To verify this, the one- and two-step detections for glucose were performed. As is known, natural glucose oxidase has an optimum pH of 5.5.<sup>53</sup> Hence, a compromised pH value of 6.0 was adopted in the one-step detection, yielding a linear range of 0.05–1.0 mM with a detection limit of 43.0  $\mu\text{M}$  (Figure 5a). The two-step detection shows a linear range of 0.02 to 1.5 mM and a detection limit of 19.4  $\mu\text{M}$  (Figure S27b). Possible interfering substances including fructose, maltose, lactose, and sucrose were selected to evaluate the selectivity, producing negligible signals (Figure 5b). The anti-interference ability of Mn-PC was also verified by the satisfactory recovery in assaying the spiked 0.5% serum samples (Table S4).

In addition to the spectrochemical detection, the electrochemical detection for  $\text{H}_2\text{O}_2$  and glucose was also implemented owing to the constitutional electronic structure of Mn-PC preserves the total conductivity. The reduction current increased with the increase of  $\text{H}_2\text{O}_2$  concentration (Figure 5c) and exhibited perfect linearity at  $-0.45$  V ranging from 0.1 to 10 mM (Figure 5c, inset). On the contrary, the cathodic current decreased with the increase in glucose concentration because of the consumption of  $\text{O}_2$  in course of glucose oxidization (Figure 5d), consistent with the previously reported results.<sup>23,54</sup> A linear range from 0.1 to 1 mM was obtained (Figure 5d, inset). The modified electrodes exhibited excellent stability with undiminished current after the storage at 4 °C for 1 month (Figure 5e,f). Additionally, similar tendency in current change was re-produced when adding 10 times concentrated interferences during glucose detection, suggesting the anti-interference properties of this electrochemical biosensor (Figure 5g).

## CONCLUSIONS

In short, the similarity between metalloporphyrins and the active center of many natural enzymes gives us the biomimetic implications for design of the Mn-PC structures. It inherits the merits of single-atom and MOF-based nanozymes in maximized utilization of metal atoms and meanwhile has the unique superiority in terms of defined molecular structures, easy characterization, and high yield. This contributes to the precise calculation for the  $K_{\text{cat}}$  value of every single active site in the real sense. The super-consistent microenvironment of the metal atoms endows the Mn-PC with a super long-term stability and exclusive POD-like activity with neither OXD-like nor CAT-like activity. The elaborate investigation for the



**Figure 5.** (a) Linear range and (b) selectivity of glucose detection. (c) Progressive cyclic voltammograms at Mn-PC/GCE in 10 mM pH 7.0 PBS containing different concentrations of  $\text{H}_2\text{O}_2$  (Inset: Linear range of electrocatalytic current of  $\text{H}_2\text{O}_2$  vs its concentrations at a potential of  $-0.45$  V). (d) Progressive cyclic voltammograms at Mn-PC@GOD/GCE in 10 mM pH 7.0 PBS containing different concentrations of glucose (Inset: Linear range of electrocatalytic current of glucose vs its concentrations at a potential of  $-0.45$  V). The stability of Mn-PC/GCE and Mn-PC@GOD/GCE for detection of 5 mM  $\text{H}_2\text{O}_2$  (e) and 0.1 mM glucose (f), respectively. (g) Amperometric time curve at Mn-PC@GOD/GCE in 10 mM pH 7.0 PBS upon successive injection of glucose (0.1 mM) or  $\text{H}_2\text{O}_2$  (0.1 mM) for each step at  $-0.48$  V. The black arrows stands for the addition of four interferents (1.0 mM) successively.

source of POD activity including the hydroxyl radicals scavenging assay and ESR spectroscopy have ensured the hydroxyl radical involved mechanism in which the colorimetric reaction would occur without the need for any other stimulus such as heat or photoirradiation. The colorimetric biosensor for  $\text{H}_2\text{O}_2$ , sarcosine, and glucose was established via a one-step strategy, which has convenience in the detection process and shows a respective detection limit of 1.9, 2.3, and 43.0  $\mu\text{M}$ . Meanwhile, electrochemical detection of glucose and  $\text{H}_2\text{O}_2$  is also demonstrated. It provides a new design philosophy for the nanozymes by locating the singly dispersed active sites in consistent microenvironment through supramolecular self-assembly, whose architecture can be simply adjusted and altered by rational design of the supramolecule configuration.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.2c03206>.

Materials and instruments, synthesis of PC, Zn-PC, and Mn-PC, the POD, OXD, and CAT-like activity of PC, Zn-PC, and Mn-PC, schematic illustration for the synthetic route of PC, Zn-PC, and Mn-PC,  $^1\text{H}$  NMR,  $^{31}\text{P}$  NMR,  $^{13}\text{C}$  DEPTQ NMR, and 2D COSY/DOSY NMR spectra of PC/Zn-PC/Mn-PC, measured/calculated isotope patterns for different charge states observed from PC/Zn-PC/Mn-PC, HRTEM image of Mn-PC, UV-vis absorbance spectra of the OPD/ABTS-Mn-PC system, UV-vis spectra of Mn-PC (or Mn-PM) incubated with  $\text{H}_2\text{O}_2$ /TMB, UV-vis spectra of different concentrations of Mn-PC incubated with  $\text{H}_2\text{O}_2$ , TMB, and thiourea, optimization for the conditions of the enzymatic reaction (including pH, temperature, reaction time, and concentration of Mn-PC), Michaelis-Menten plots, Zeta potential comparison of TMB and Mn-PC, Lineweaver-Burk plots for various concentrations of TMB and  $\text{H}_2\text{O}_2$ , colorimetric detection of  $\text{H}_2\text{O}_2$ , optimization of pH in sarcosine detection, colorimetric detection of sarcosine/glucose via a two-step method, comparison of the kinetic parameters of Mn-PC, HRP, and some other nanozymes, and recovery rate of  $\text{H}_2\text{O}_2$  in lens care solution, sarcosine in 4% human urine, and glucose in 0.5% human serum (PDF)

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## Author Contributions

<sup>||</sup>C.Z. and Y.Z. contributed equally to this work.

## Author Contributions

Y.D. and M.W. conceived the work. Y.T.Z. synthesized and characterized the structure of supramolecular nanocages. C.Y.Z. and Z.M.S. studied the enzyme-like activities of three supramolecular nanocages and investigated the sensor performances. Y.Z., C.Y.Z., Y.T.Z., and Q.Y.L. contributed to the data analyses. Y.D., M.W., Y.Z., C.Y.Z., and Y.T.Z. drafted the manuscript.

## Notes

The authors declare no competing financial interest.

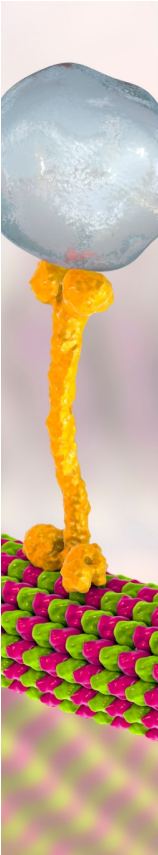
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